

DIGITAL LOGIC & COMPUTER ARCHITECTURE

8-Bit Signed Binary Representations

Design & Simulation using Sign-Magnitude, 1's Complement, and 2's Complement with
Logic Gates, Timers, and Hex Displays

Unsigned vs. Signed Numbers

Unsigned Binary

Represents strictly non-negative values from `0` to `255` ($2^n - 1$).

- All 8 bits are dedicated to magnitude.
- Value equation: $\text{Sum}(b_i * 2^i)$ for $i = 0..7$.
- Cannot represent negative values or directional data.

Signed Binary Formats

Allocates the Most Significant Bit (MSB, Bit 7) as the **Sign Bit**.

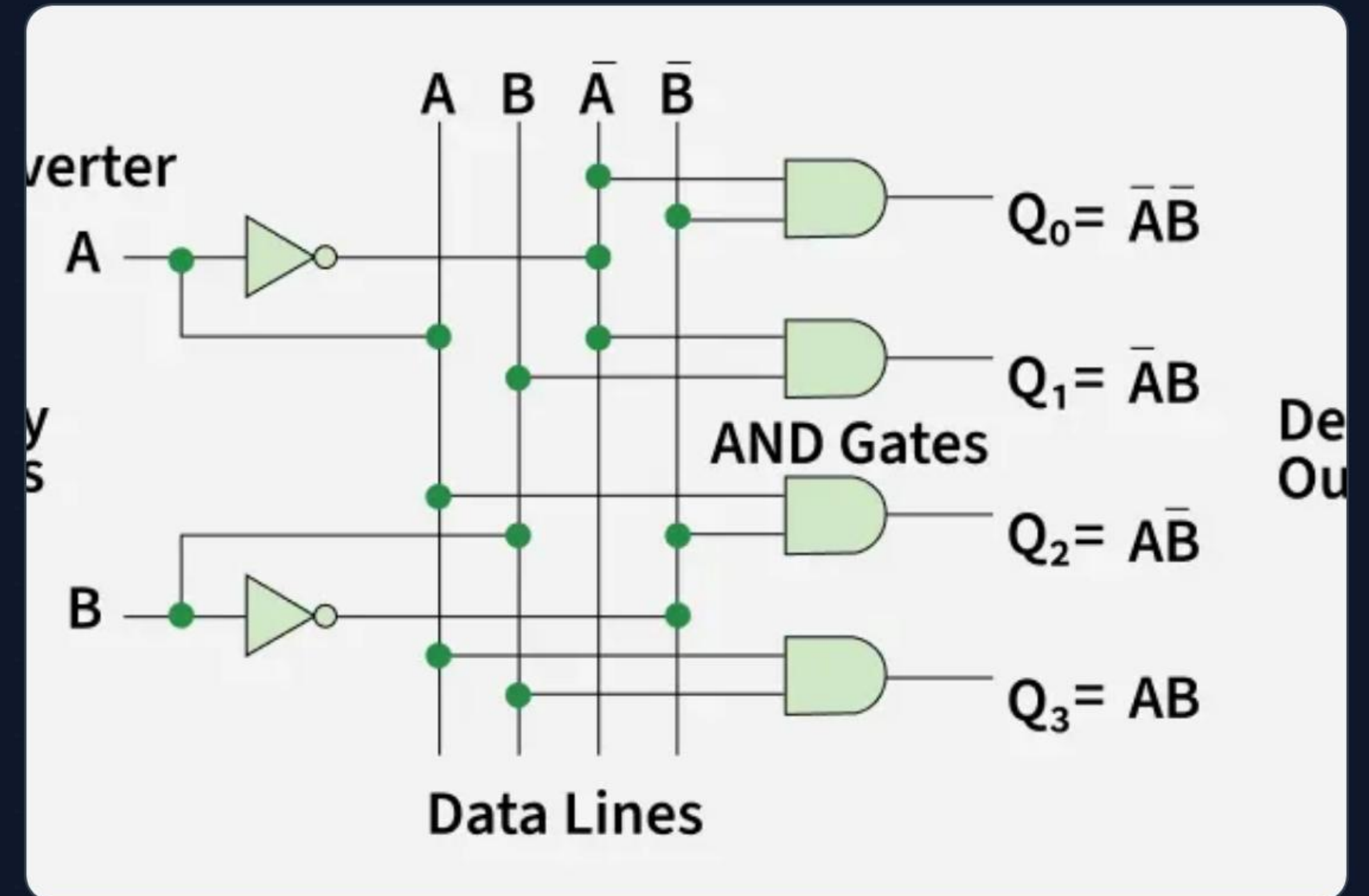
- $\text{MSB} = 0$ indicates a positive number.
- $\text{MSB} = 1$ indicates a negative number.
- Three primary schemes: Sign-Magnitude, 1's Complement, 2's Complement.

1. Sign-Magnitude Format

Structure & Characteristics

The simplest conceptual representation splitting an 8-bit word into sign and magnitude components:

- **Bit 7 (MSB):** Sign Bit ($0 = +$, $1 = -$).
- **Bits 0-6:** True magnitude in binary.
- **Range:** -127 to $+127$ ($-(2^7 - 1)$ to $+(2^7 - 1)$).
- **Limitation:** Dual representation of zero ($+0 = 00000000$ and $-0 = 10000000$), causing hardware design redundancy.



2. 1's Complement & NOT Gate Logic



Bit Inversion

Negative numbers are formed by inverting every single bit of the positive number representation ($0 \rightarrow 1$ and $1 \rightarrow 0$).



NOT Gate Circuitry

Hardware implementation relies on 8 parallel Inverter (NOT) gates (e.g., IC 7404) driven by a conditional inversion signal.



End-Around Carry

Arithmetic operations require an extra step: any carry-out from the MSB during addition must be added back to the LSB.

3. 2's Complement Representation

256

Unique Values (-128 to +127)

Standard Industry Format

Obtained by taking the 1's complement of a binary number and adding 1 to the Least Significant Bit (LSB).

Mathematical Formula:

$$\text{2's Complement} = (\text{1's Complement}) + 1$$

- Eliminates the double-zero anomaly (00000000 is uniquely zero).
- Allows subtraction to be performed using standard addition hardware.

Comparison Matrix (8-Bit Formats)

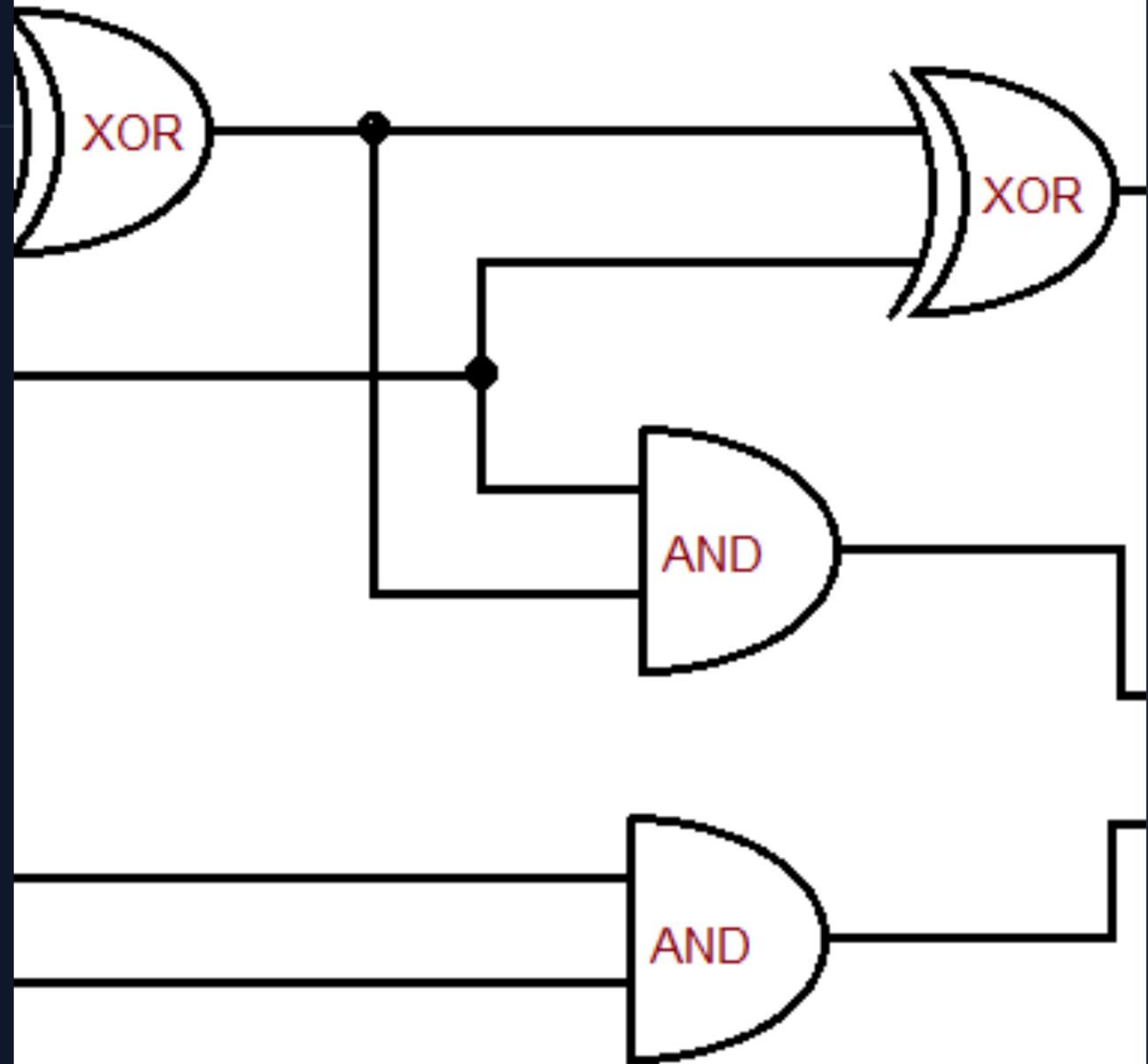
Decimal Value	Sign-Magnitude	1's Complement	2's Complement
+25	0001 1001	0001 1001	0001 1001
-25	1001 1001	1110 0110	1110 0111
+0	0000 0000	0000 0000	0000 0000
-0 / Extra	1000 0000 (-0)	1111 1111 (-0)	1000 0000 (-128)

Hardware Architecture

XOR & Ripple Adder Logic

A unified circuit transforms positive numbers into 2's complement on demand:

1. **XOR Array (7486):** Acts as controlled bit inverters when `SUB = 1`.
2. **Full Adder (7483 / 74283):** Adds the carry-in (`C_in = 1`) to complete the 2's complement calculation instantly.



Simulation Output: Hex Displays



Nibble Splitting

An 8-bit output (`D7 .. D0`) is split into two 4-bit nibbles: Upper Hex (`D7 .. D4`) and Lower Hex (`D3 .. D0`).



Decoder ICs (7447/74147)

Translates binary nibble signals into 7-segment LED driver currents to render hex digits (`0x00` to `0xFF`).



Sign Status Indicator

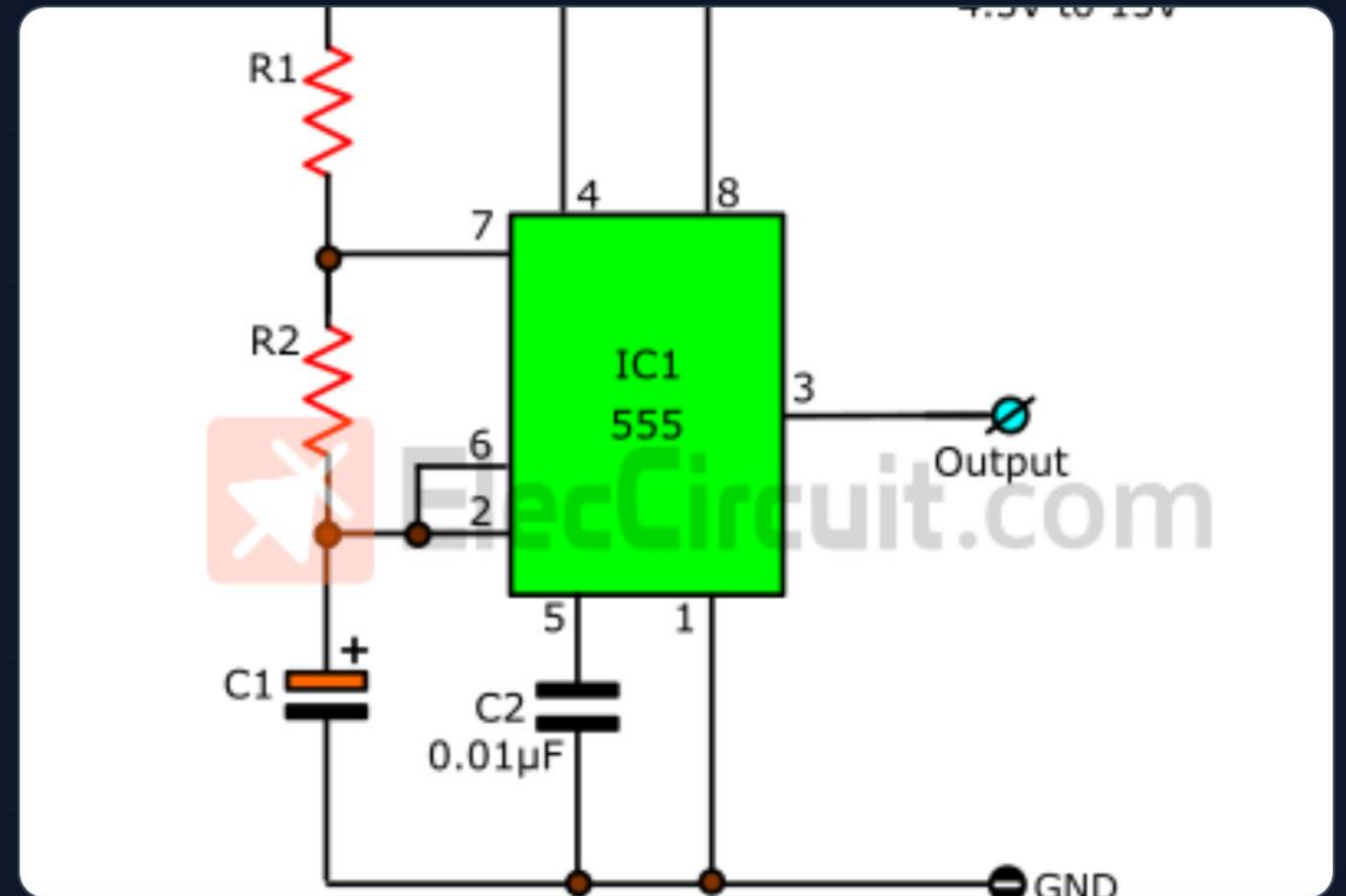
A dedicated LED connected directly to the MSB (`Bit 7`) signals overall number polarity (`OFF = Positive` , `ON = Negative`).

Automated Simulation & Timers

Clocking with 555 Timer IC

To verify the design, test vectors are dynamically fed using an astable 555 timer pulse generator:

- **555 Timer in Astable Mode:** Generates continuous square wave clock pulses.
- **Binary Counter (7493):** Cycles through sequential 8-bit input states on clock transitions.
- **Oscilloscope Probe:** Captures switching speed and propagation delay across logic gates.

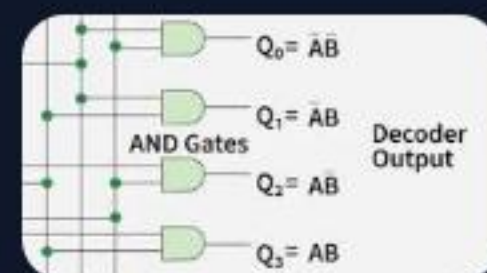


Questions & Discussion

Design & Simulation of 8-Bit Signed Binary Representation Complete

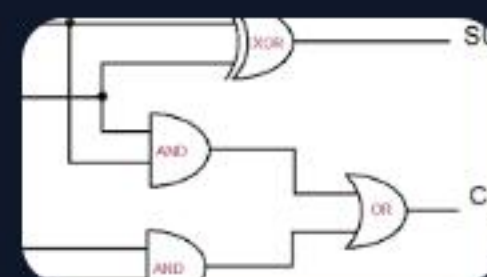
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Image Sources



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